

A Dictionary-Based Fuzzy Search and Spell Correction Engine for Smart Shopping

```
PS C:\Users\akshi\Downloads\2nd year 1st sem\Smart-cart> java -cp out
Dictionary loaded with 1749 words.
Catalog loaded with 434 products.

===== SmartCart: Shopping Assistant =====
1. Multi-Keyword Product Filter (Aho-Corasick)
2. Exact Phrase Search in Catalog (KMP)
3. Typo-Tolerant Word Search (Wagner-Fischer + Damerau-Levenshtein)
4. Autocomplete / Suggest-as-you-type
5. Related Products ("You might also like")
6. View / Manage Cart
7. View Recent Search History
0. Exit
```

1. Multi-Keyword Product Filter – Aho-Corasick

This feature allows users to search for products using multiple keywords at the same time. The Aho-Corasick algorithm searches the product catalog for all the entered keywords and displays the matching products.

Screenshot:

Enter choice: 1

Catalog has 434 products. Showing first 15:

- seiko washingmachine childproof gold
- vivo decoder tearresistant leather
- steelseries lamp foldable zinc
- haier earbuds durable fiberglass
- noise keyboard noisecancelling metal
- karbonn webcam foldable glass
- godrej cctv recyclable rose
- letv lamp childproof gold
- godrej headphones flameretardant carbon
- casio cabinet smart glass
- letv dock flameretardant glossy
- acer lamp odorresistant transparent
- coolermaster router shatterproof gray
- poco mouse biodegradable yellow
- nothing waterheater tearresistant acrylic

How many keywords do you want to filter by? 3

Keyword 1: cctv

Keyword 2: phone

Keyword 3: cam

```

--- Matching Products ---
"karbonn webcam foldable glass" -> matched keywords: [cam]
"godrej cctv recyclable rose" -> matched keywords: [cctv]
"godrej headphones flameretardant carbon" -> matched keywords: [phone]
"sharp camera quickdry rose" -> matched keywords: [cam]
"redmi dashcam affordable nylonmesh" -> matched keywords: [cam]
"sennheiser camera multipurpose gold" -> matched keywords: [cam]
"acer smartphone stainresistant steel" -> matched keywords: [phone]
"steelseries actioncamera automatic cast iron" -> matched keywords: [cam]
"tecno actioncamera handmade stainlesssteel" -> matched keywords: [cam]
"oppoenco dashcam retina wood" -> matched keywords: [cam]
"realme actioncamera classic cast iron" -> matched keywords: [cam]
"hisense smartphone affordable steel" -> matched keywords: [phone]
"viewsonic securitycamera mechanical chrome" -> matched keywords: [cam]
"gopro headphones affordable glossy" -> matched keywords: [phone]
"evga actioncamera imported navy" -> matched keywords: [cam]
"campingchair artisanal" -> matched keywords: [cam]

```

2. Exact Phrase Search – KMP

This feature searches the product catalog for an exact phrase entered by the user. The KMP algorithm efficiently identifies whether the phrase exists and displays the matching product and position.

Screenshot:

```

===== SmartCart: Shopping Assistant =====
1. Multi-Keyword Product Filter (Aho-Corasick)
2. Exact Phrase Search in Catalog (KMP)
3. Typo-Tolerant Word Search (Wagner-Fischer + Damerau-Levenshtein)
4. Autocomplete / Suggest-as-you-type
5. Related Products ("You might also like")
6. View / Manage Cart
7. View Recent Search History
0. Exit
=====
Enter choice: 2

```

```

Enter an exact phrase to search for: phone
"godrej headphones flameretardant carbon" -> found at index(es) [11]
"acer smartphone stainresistant steel" -> found at index(es) [10]
"hisense smartphone affordable steel" -> found at index(es) [13]
"gopro headphones affordable glossy" -> found at index(es) [10]

```

3. Typo-Tolerant Word Search

This feature helps users when they make spelling mistakes while searching. Wagner-Fischer and Damerau-Levenshtein algorithms compare the entered word with dictionary terms and provide the closest correction.

Screenshot:

```

===== SmartCart: Shopping Assistant =====
1. Multi-Keyword Product Filter (Aho-Corasick)
2. Exact Phrase Search in Catalog (KMP)
3. Typo-Tolerant Word Search (Wagner-Fischer + Damerau-Levenshtein)
4. Autocomplete / Suggest-as-you-type
5. Related Products ("You might also like")
6. View / Manage Cart
7. View Recent Search History
0. Exit
=====
Enter choice: 3

```

```

Search for a word (e.g. a mistyped brand/category): woodn
No exact match for "woodn".
Did you mean one of these?
1. "wood" (Wagner-Fischer: 1, Damerau-Levenshtein: 1)
2. "wool" (Wagner-Fischer: 2, Damerau-Levenshtein: 2)
3. "boots" (Wagner-Fischer: 3, Damerau-Levenshtein: 3)

```

4. Autocomplete / Suggest-as-you-type

This feature provides word suggestions based on the characters entered by the user. It uses the shopping dictionary to suggest relevant product or brand terms.

Screenshot:

```

===== SmartCart: Shopping Assistant =====
1. Multi-Keyword Product Filter (Aho-Corasick)
2. Exact Phrase Search in Catalog (KMP)
3. Typo-Tolerant Word Search (Wagner-Fischer + Damerau-Levenshtein)
4. Autocomplete / Suggest-as-you-type
5. Related Products ("You might also like")
6. View / Manage Cart
7. View Recent Search History
0. Exit
=====
Enter choice: 4

Start typing a word (prefix): gam
Suggestions for "gam":
- gamepad
- gaming

```

5. Related Products – “You might also like”

This feature displays products related to the selected or searched product. It helps users discover additional products from the catalog.

Screenshot:

```
===== SmartCart: Shopping Assistant =====
1. Multi-Keyword Product Filter (Aho-Corasick)
2. Exact Phrase Search in Catalog (KMP)
3. Typo-Tolerant Word Search (Wagner-Fischer + Damerau-Levenshtein)
4. Autocomplete / Suggest-as-you-type
5. Related Products ("You might also like")
6. View / Manage Cart
7. View Recent Search History
0. Exit
=====
Enter choice: 5
```

```
Catalog has 434 products. Showing first 15:
- seiko washingmachine childproof gold
- vivo decoder tearresistant leather
- steelseries lamp foldable zinc
- haier earbuds durable fiberglass
- noise keyboard noisecancelling metal
- karbonn webcam foldable glass
- godrej cctv recyclable rose
- letv lamp childproof gold
- godrej headphones flameretardant carbon
- casio cabinet smart glass
- letv dock flameretardant glossy
- acer lamp odorresistant transparent
- coolermaster router shatterproof gray
- poco mouse biodegradable yellow
- nothing waterheater tearresistant acrylic

Enter a product (or part of one) to get recommendations for: keyboard
Because you looked at: "noise keyboard noisecancelling metal"
You might also like:
- noise nebulizer stackable cream
- toshiba keyboard slim porcelain
- hp inc oximeter curved metal
- sandisk alarm heavyduty metal
- noise decoder stretchable titanium
```

6. View / Manage Cart

This feature allows users to view and manage the products added to their shopping cart. It provides a basic shopping-cart experience within SmartCart.

Screenshot:

```
--- Cart Menu ---
1. Add product to cart
2. Remove product from cart
3. View cart
4. Clear cart
0. Back to main menu
Enter choice: 1
Enter a product (or part of one) to add: noise decoder
Quantity: 1
Added "noise decoder stretchable titanium" to cart.
```

```
0. Back to main menu
Enter choice: 3

--- Your Cart ---
1. noise decoder stretchable titanium (qty: 1)
Total items: 1
```

```
--- Cart Menu ---
1. Add product to cart
2. Remove product from cart
3. View cart
4. Clear cart
0. Back to main menu
Enter choice: 0
```

7. Recent Search History

This feature stores the user's recent searches and displays them when requested. It allows users to review their previous search queries.

Screenshot:

```
===== SmartCart: Shopping Assistant =====
1. Multi-Keyword Product Filter (Aho-Corasick)
2. Exact Phrase Search in Catalog (KMP)
3. Typo-Tolerant Word Search (Wagner-Fischer + Damerau-Levenshtein)
4. Autocomplete / Suggest-as-you-type
5. Related Products ("You might also like")
6. View / Manage Cart
7. View Recent Search History
0. Exit
```

```
=====
```

```
Enter choice: 7
```

```
--- Recent Searches (most recent first) ---
```

```
1. [recommend] "keyboard"
2. [autocomplete] "gam"
3. [typo-search] "woodn"
4. [phrase-search] "phone"
5. [keyword-filter] "cctv, phone, cam"
```